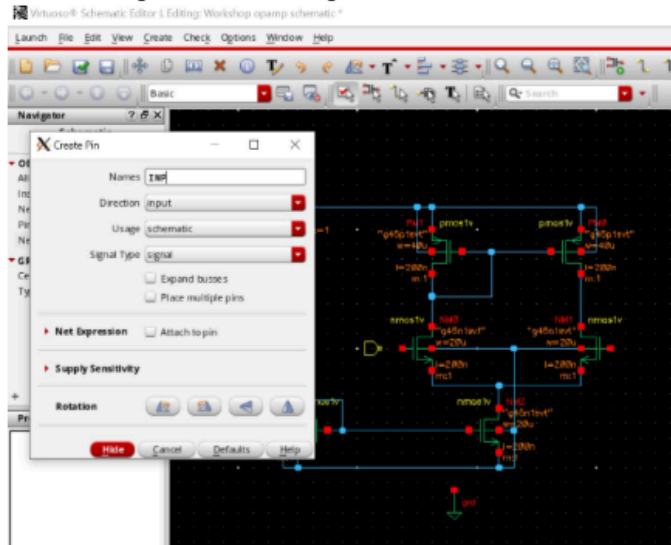
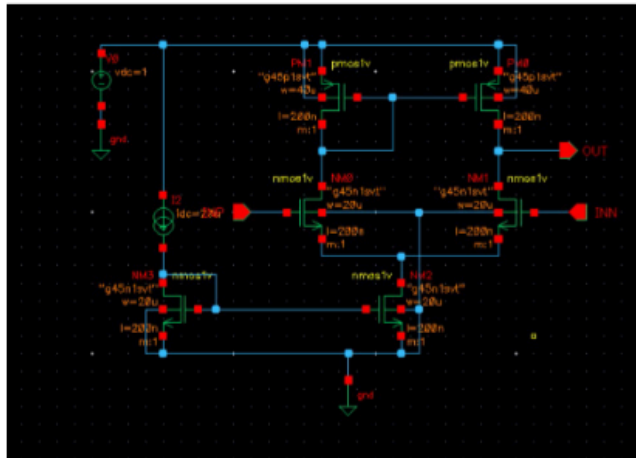


Opamp Create Pin



- Follow the earlier workshop, create a simple opamp schematic as shown in the figure
- Type “p” to create pin
- Type “INP” for “Names” and select “Direction” as “input”
- Repeat the steps to create “INN” (input) and “OUT” (output) as shown in the bottom figure

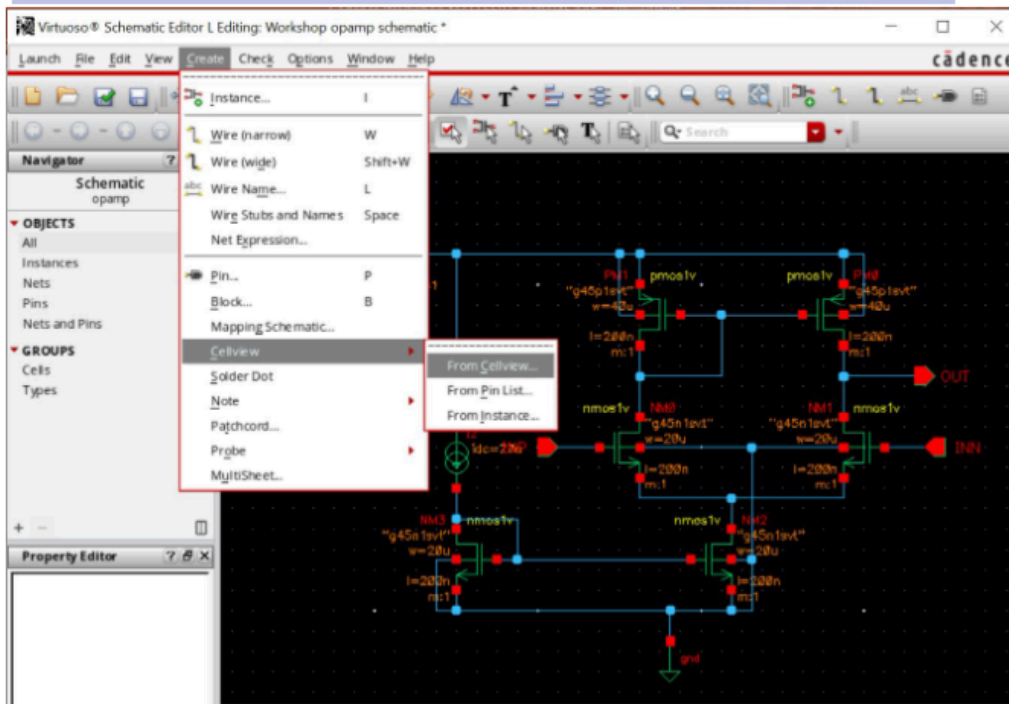
1. All NMOS sizing is $20\mu/200n$
2. All PMOS sizing is $40\mu/200n$
3. DC voltage is 1V
4. DC current is $20\mu A$



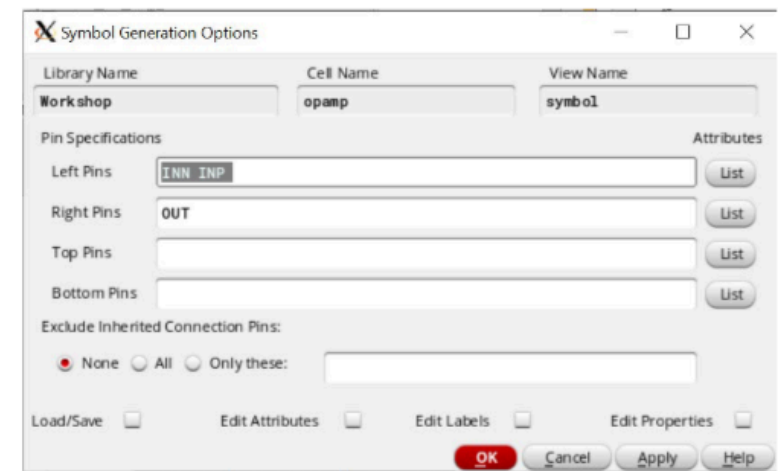
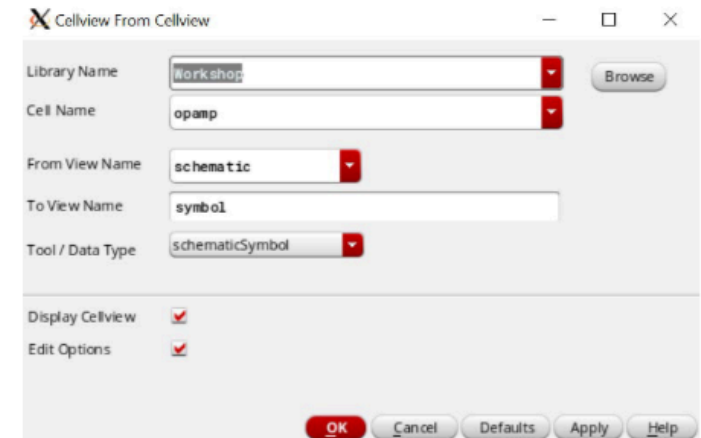
- to bring fabrication in
 - the VDD, GND and Vbias should be pins
 - so the block is totally abstracted
- Supply use bidirectional
 - then in/out

Opamp Create Symbol

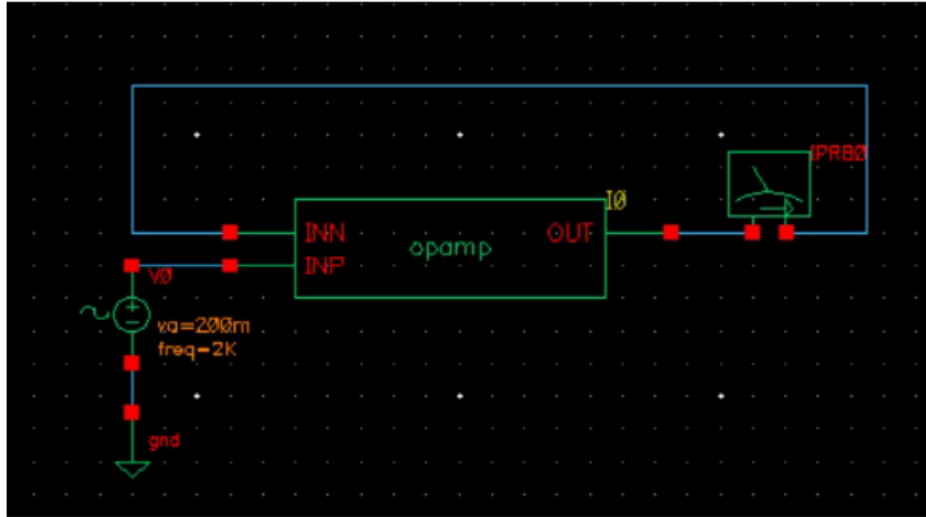
Opamp Create Symbol



- Select **Create->Cellview->From Cellview**, “Cellview” window will pop up
- Click “OK” and “Symbol Generation” window will pop up
- You can decide the PIN orientation
- The resulting symbol is shown in the bottom right figure
 - draw the symbol better for better hierarchy design



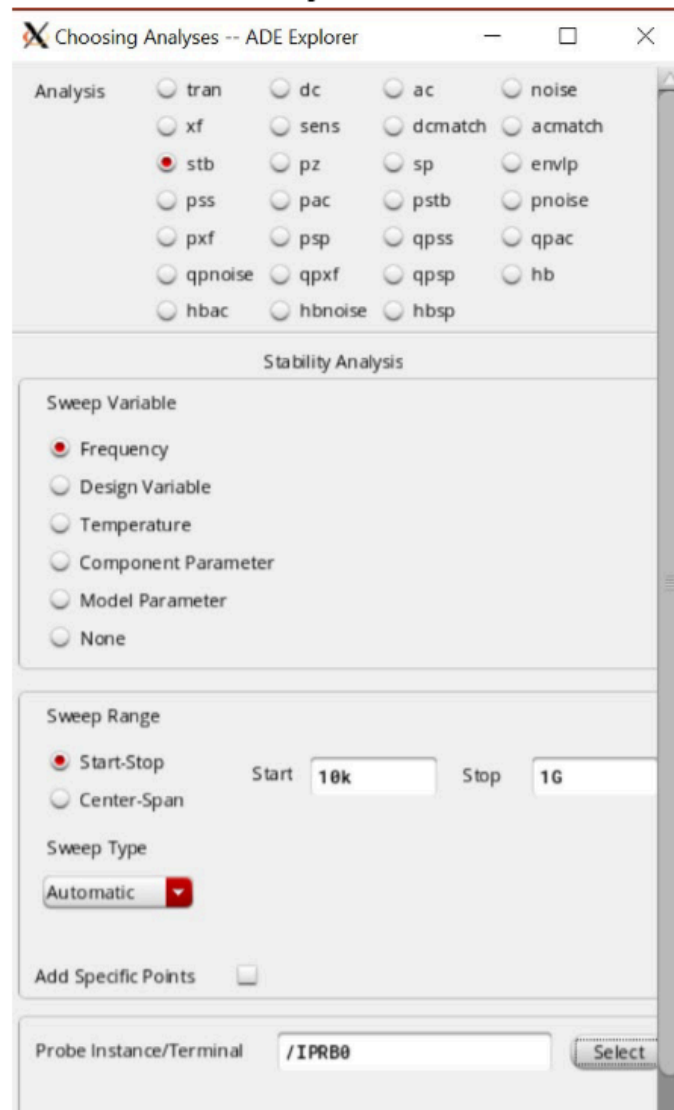
Create Unity Gain Amplifier



- Create non-inverting amplifier by using components from analogLib and Workshop libraries
- There are two new components:
 1. **vsin**: Sinusoidal source with DC of 0.5V, amplitude of 0.2V and frequency of 2kHz. Also set the “AC Magnitude” to 1
 2. **iprobe**: Useful for stability analysis (Bode plot and phase margin). It should be placed at a location that will break the feedback loop

- the AC magnitude is for AC simulation
 - set to 1V, the voltage output = ac gain
 - AC simulation is already linearised based on DC operation point, so 1V doesn't matter for small signal concerns
- DC is half supply
- iprobe
 - it will break the loop properly
 - to do it manually, is very troublesome, since the output needs to be loaded to the INN of another, so need another dummy op-amp to give the load
 - if not done properly, the result cannot be trusted
 - iprobe takes care of it nicely
- unity gain means the loop gain has feedback factor of 1

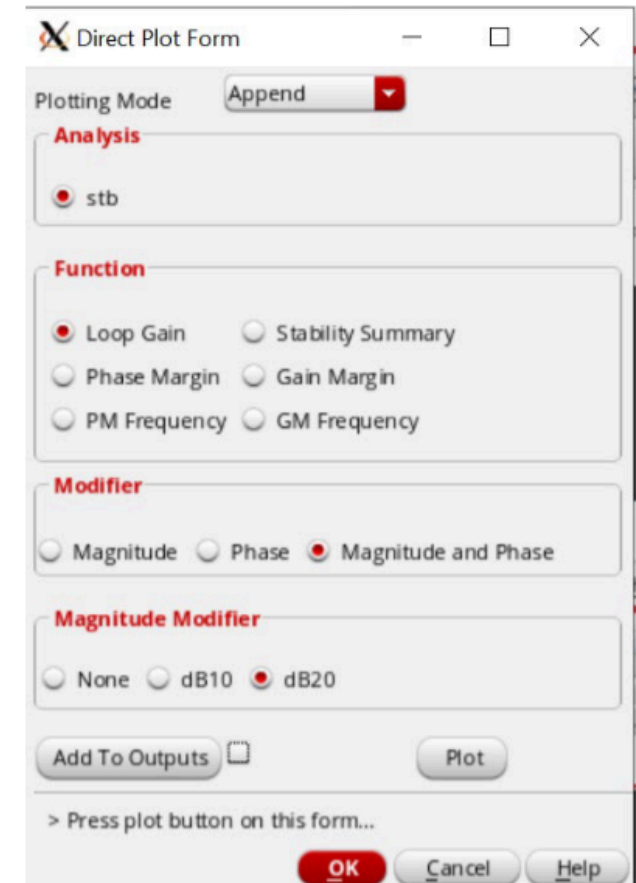
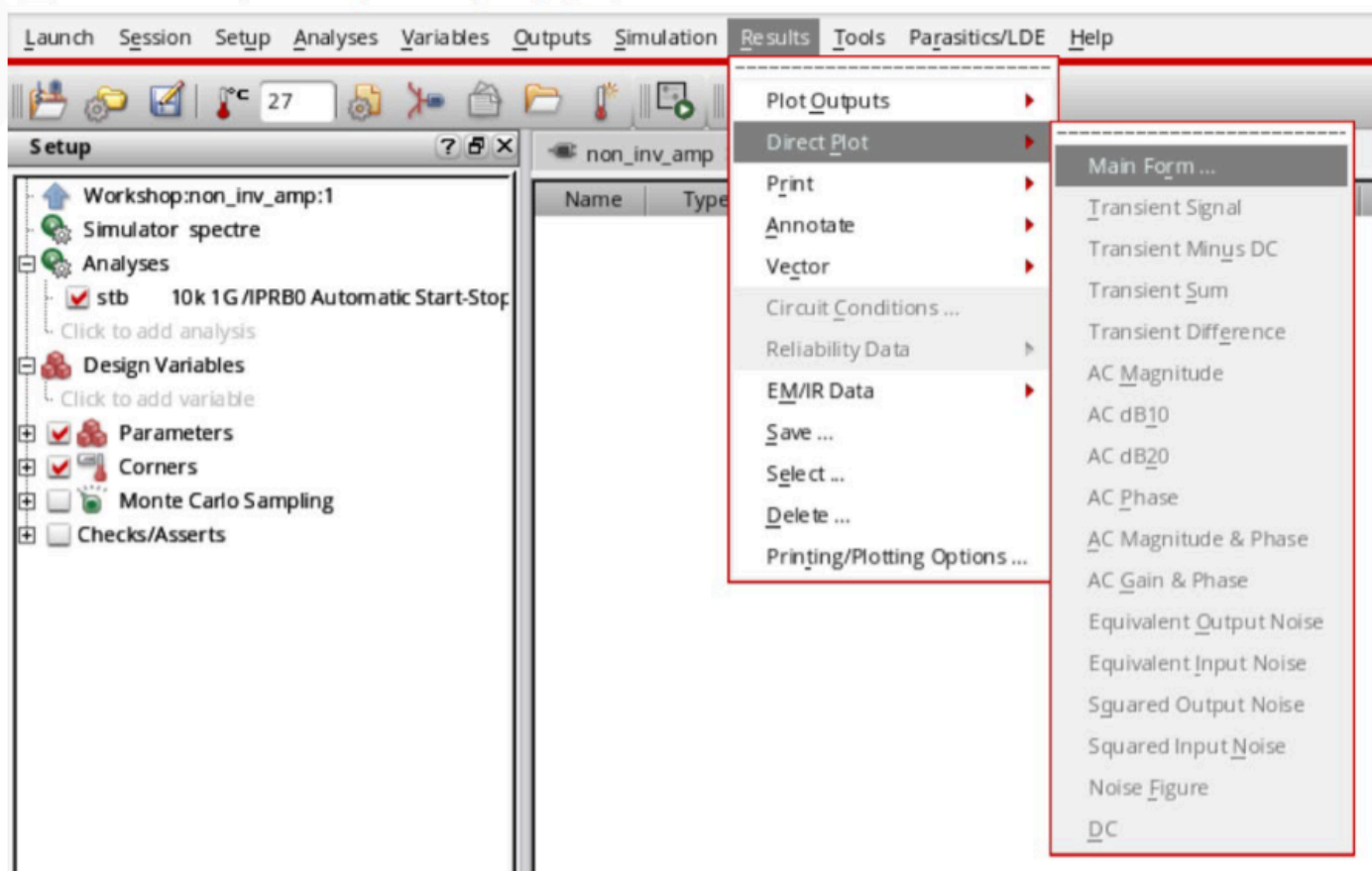
Stability Simulation



- Refer to earlier slide for model library setup and choose stability analysis
- Choose “stb” under “Analysis”
- Select “Frequency” under “Sweep Variable”
- Under “Sweep Range”, set “Start” to “10k” and “Stop” to “1G”
- For “Probe Instance/Terminal”, type “/IPRB0” (this is the iprobe instance in the schematic, you can also click “Select” and directly choose from schematic)
- The good thing of using iprobe is to enable loop gain analysis without actually breaking the loop
- Run simulation by selecting **Simulation-> Netlist and Run**

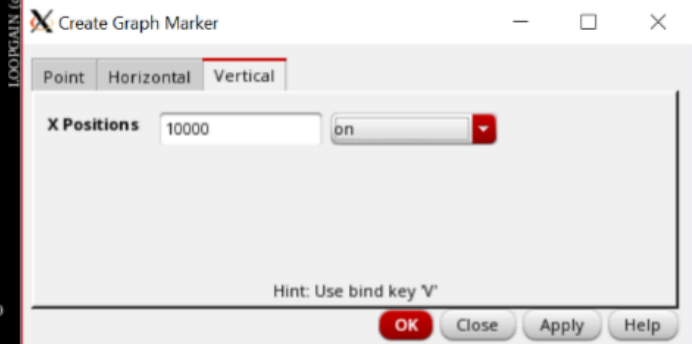
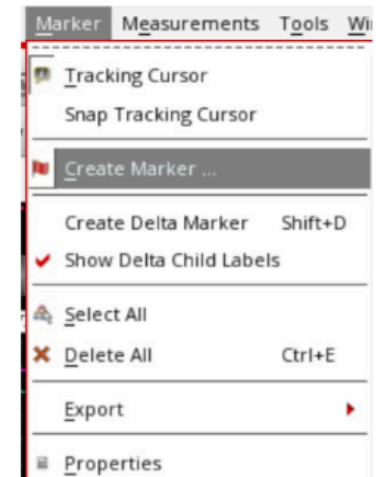
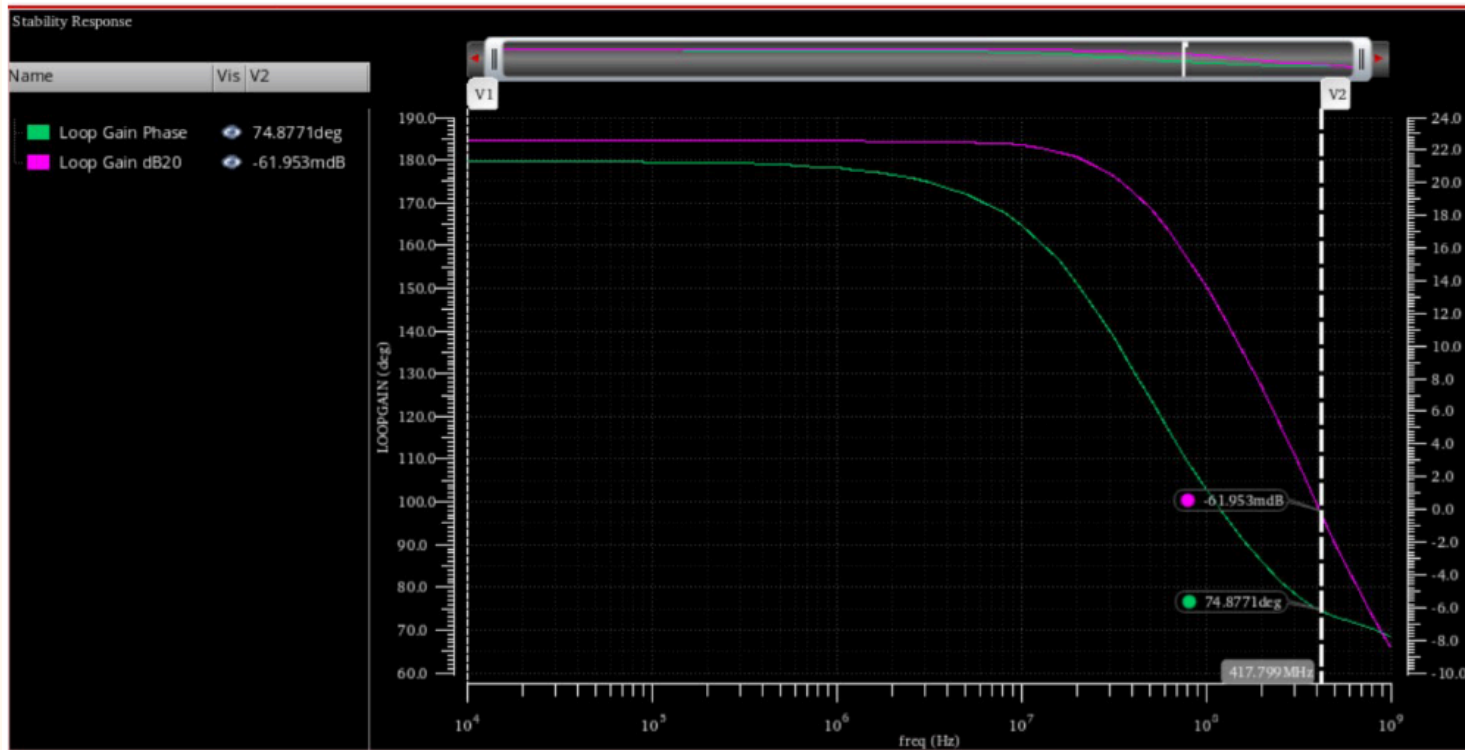
Stability Plot

VirtualSPICE - PCB Explorer Editing: Workshop:non_inv_amp:measure



- Choose **Results->Direct Plot->Main Form**, “Direct Plot” window pop up
- Choose “Loop Gain” under Function, “Magnitude and Phase” under “Modifier”, and dB20. Click “Plot”.

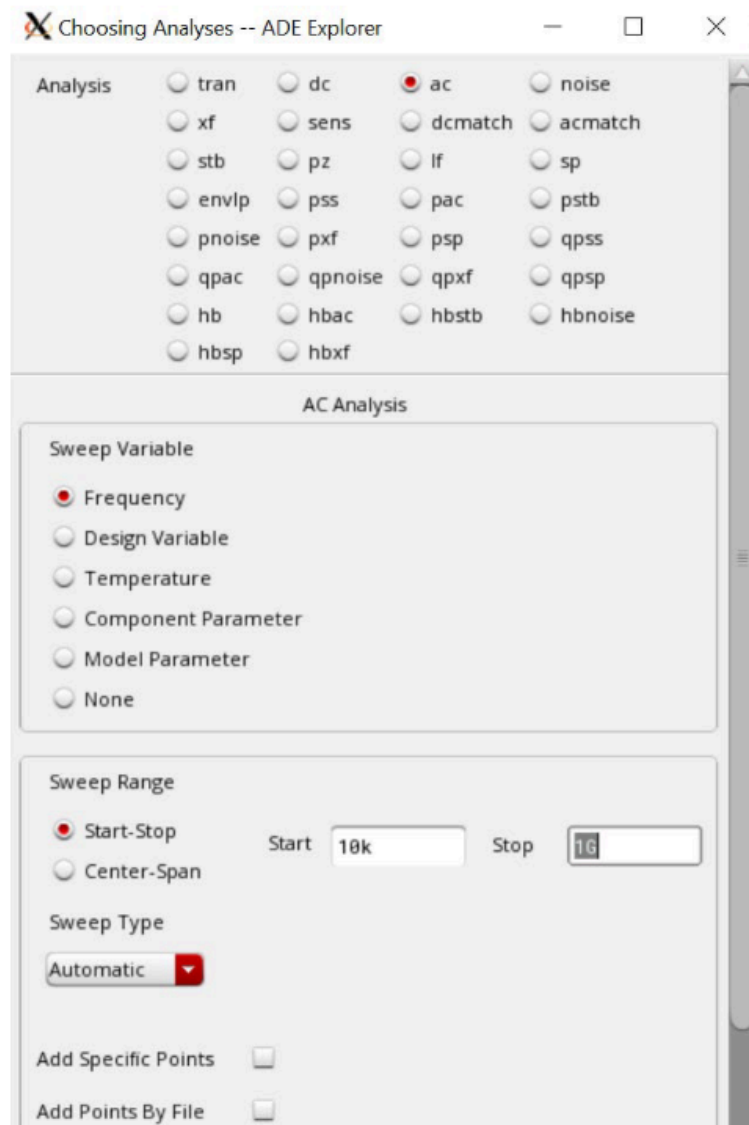
Stability Result



- Choose **Marker->Create Marker**, “Create Graph Marker” window pop up
- Choose “Vertical” and select “On”
- Type “v” on the plot to show the cursor
- Read off the phase value at frequency when the loop gain drop to 0 dB, that will be your phase margin
- Note that such loop gain plot has taken the feedback factor into account

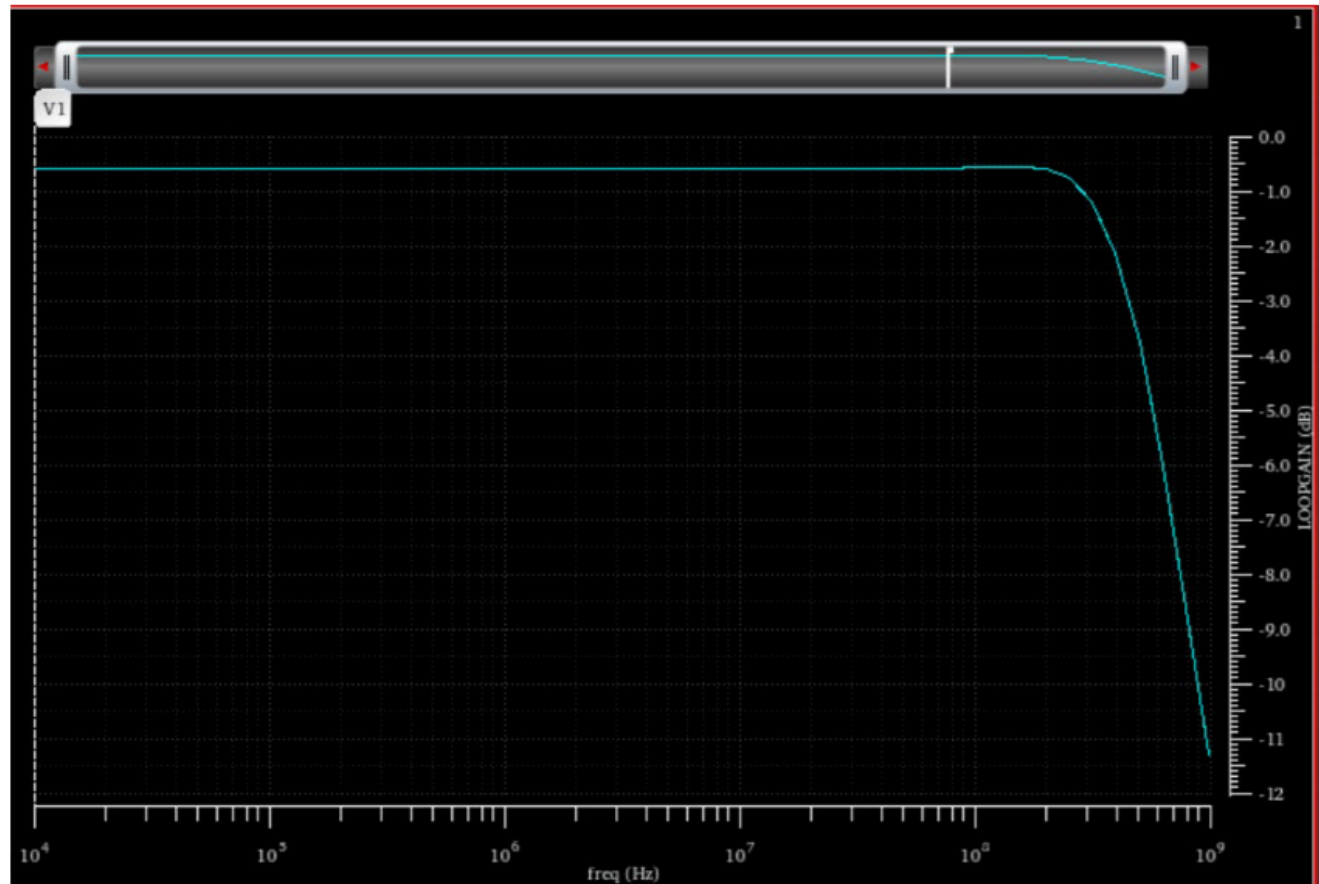
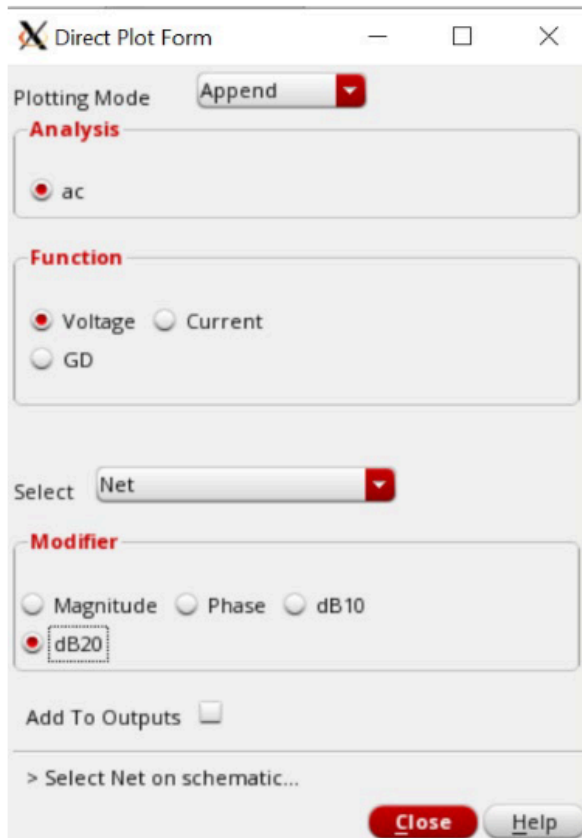
- for the interpretation of the Bode plot in cadence and why its not the same as the conventional one
 - see [here](#)

AC Simulation



- Choose “ac” under “Analysis”
- Choose “Frequency” under “Sweep Variable”
- Under “Sweep Range”, type “10k” for “Start” and “1G” for “Stop”
- Run simulation by selecting **Simulation->Netlist and Run**

AC Result

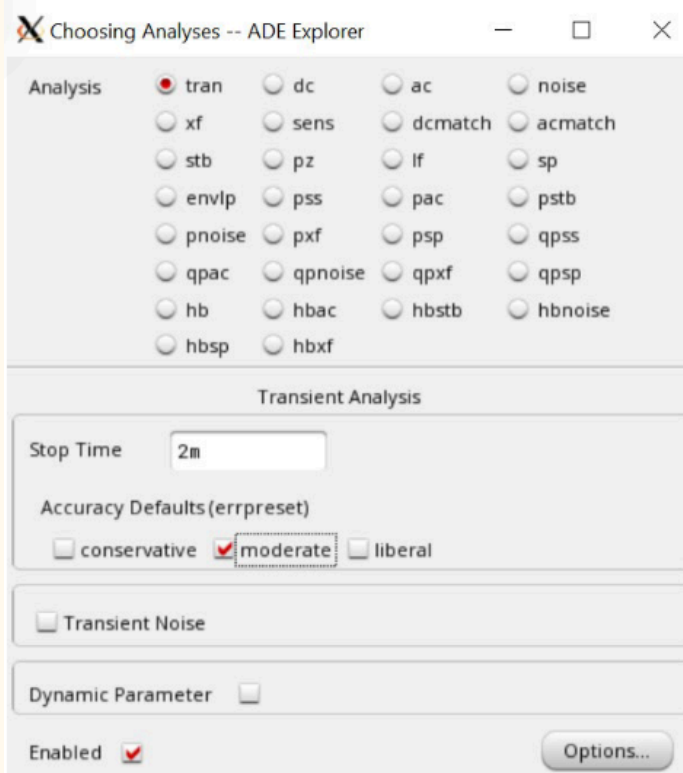


- Select **Results->Direct Plot**
- Select “Voltage” under “Function” and “Net”. Finally “dB20” under “Modifier”
- Then select the “OUT” net from the schematic

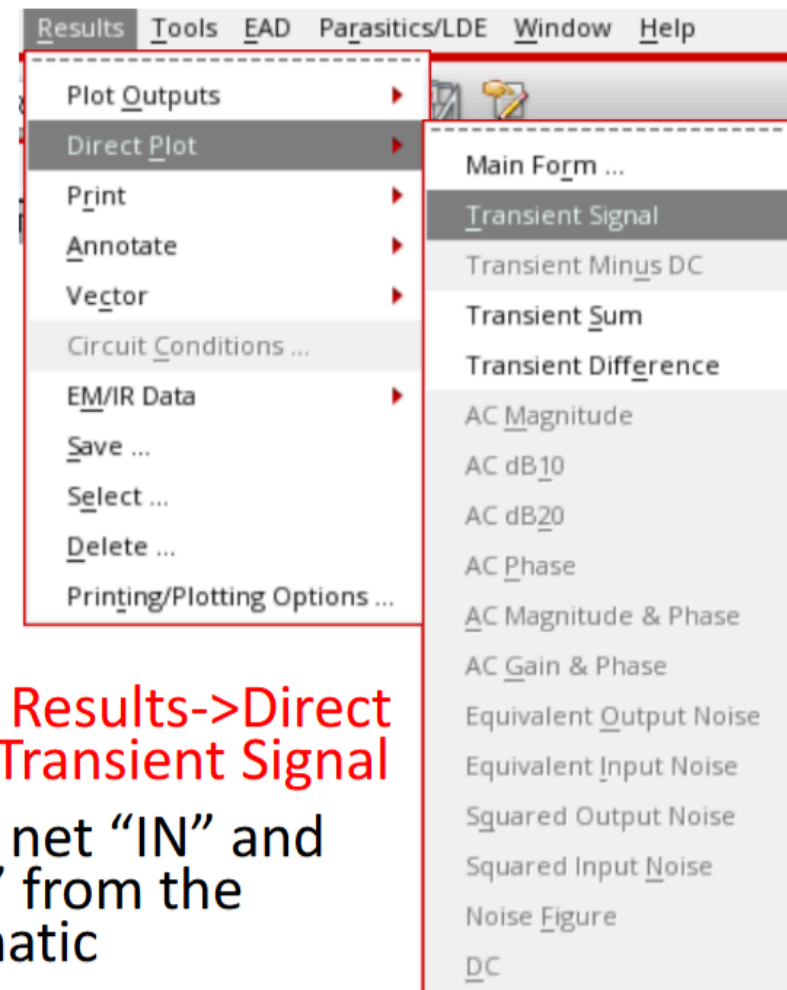
- the AC gain suppose to be 0dB
- but its not because the op amp design is anyhow one
 - the gain is not big
 - so virtual short is not there

- so got error from 0dB

Transient Simulation



- Choose “tran” under “Analysis”
- Under “Stop Time” type “2m”
- Choose “moderate” under “Accuracy”
- Run simulation by selecting **Simulation->Netlist and Run**



- Select **Results->Direct Plot->Transient Signal**
- Select net “IN” and “OUT” from the schematic

Transient Result

